

CRP HEL and Wetland Screening Tool

(Prototype)

Methodology, Formula Reference & Data Sources

Based on: USDA NRCS SSURGO Database, NRCS FOTG R-Factors, NOAA Climate Data Online API, USGS 3DEP Elevation Data, NLCD Land Cover Classification, NHD/NWI Wetland Database

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Status: Results are indicative only — not an official RUSLE2 or HEL determination

WETLAND DETERMINATION METHODOLOGY (UPDATED)

The tool now includes a comprehensive 4-indicator wetland assessment per the Federal Interagency Wetlands Delineation Manual. The display has been refined to show only actionable indicators.

1. Hydric Soils Detection (SSURGO hydricrating)

- Source: SSURGO hydricrating field (NRCS soil property)
- Accuracy: $\pm 5\%$ (established soil classification)
- Signal: Soils formed under prolonged saturation conditions
- Display: Always shown (both Yes and No results are actionable)

2. Hydrophytic Vegetation Detection (NLCD Land Cover) — REFINED

- Source: NLCD 2019 Land Cover (MRLC WMS GetFeatureInfo API)
- Classes: Class 90 (Woody Wetlands) and Class 95 (Emergent Herbaceous Wetlands)
- Accuracy: $\pm 5\%$ (30m resolution Landsat-derived classification)
- **Smart Display (v5.2):** Shown ONLY when detected as Yes (vegetation present). Omitted when No because field conservationists will verify actual vegetation during site visits, and NLCD's 30m satellite resolution (2019 data) may not capture 1-hectare field conditions. Negative results are not actionable and can create confusion.

3. Wetland Hydrology Detection (Water Table Estimation)

- Source: SSURGO drainage class (proxy for water table depth)
- Method: NRCS hydrologic correlation with drainage classes
- Very Poorly Drained → 10 cm water table (Strong signal)
- Poorly Drained → 25 cm water table (Strong signal)
- Somewhat Poorly Drained → 45 cm water table (Possible signal)
- Well/Moderately Well Drained → >100 cm (No signal)
- Accuracy: $\pm 20\%$ (established NRCS soil morphology standards)
- Display: Always shown (both Yes and No are actionable)

4. Proximity to Water Body Detection (NHD/NWI Database)

- Source: FWS National Wetlands Inventory (NWI) via WMS GetFeatureInfo

- Coverage: Mapped wetland polygons across all US states
- Classes: Cowardin Classification (P=Palustrine, R=Riverine, L=Lacustrine, E=Estuarine, M=Marine)
- Accuracy: $\pm 5\%$ (official USGS/FWS wetland delineation)
- Confidence Levels: Strong (P/R/L systems) | Possible (E/M systems)
- Display: Always shown (both Yes and No are actionable)

Determination Confidence Scoring:

- High Confidence (3+ indicators): Strong candidate for wetland restoration (CP23, CP28)
- Medium Confidence (2 indicators): Possible wetland restoration opportunity — field verification recommended
- Low Confidence (1 indicator): Potential wetland — further investigation needed
- No Indicators: Does not appear to be wetland-forming soil

Important: Official wetland determination requires NRCS field visit and certification by NRCS conservationist per 7 CFR Sections 12.20–12.30

UI DISPLAY ENHANCEMENTS

Detailed Wetland Indicators Table (Conservationist View)

The tool displays indicators with detailed evidence from actual API queries. The NLCD indicator now uses smart display: shown only when vegetation is detected.

Example 1: Area WITH Wetland Vegetation (Atchafalaya Basin, LA)

Indicator	Detected	Evidence	Confidence
Hydric Soils	Yes	SSURGO hydricating present	High
Hydrophytic Vegetation (NLCD)	Yes	NLCD Class 90: Woody Wetlands	High
Wetland Hydrology (Water Table)	Yes	Poorly drained (25cm estim.)	High
Proximity to Water Body	Yes	NWI: PFO1/2F - Strong signal	High

Example 2: Area WITHOUT Wetland Vegetation (Non-wetland Field)

Note: The NLCD vegetation row is *omitted* when No is detected, avoiding clutter and confusion. Field conservationists will verify vegetation directly during site visit.

Indicator	Detected	Evidence	Confidence
Hydric Soils	No	SSURGO hydricating absent	—
Wetland Hydrology (Water Table)	No	Well-drained soils (>100cm)	—
Proximity to Water Body	No	No mapped wetland within 5km	—

Benefits of Enhanced Display:

- Shows actual NLCD class numbers and vegetation types (when detected)
- Displays NWI wetland codes and hydrology signal strengths
- Provides water table depth estimates derived from drainage class
- Links evidence directly to official data sources (SSURGO, NLCD, NWI)
- Enables conservationists to verify data before field visit
- Cleaner UI: Omits non-actionable negative NLCD results (v5.2 refinement)